

Tutorial 4_summary

Tutorial 4 — Loading Effects, Random Signals, Noise & Frequency Domain

Source: Tutorial 4 with Solutions.pdf

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Corresponds to: Lecture 4 (Chapters 5–6 of Bentley)

1. Electrical Loading Effects

1.1 Core Concept

State	Description
Ideal	Sensor transmits signal to next stage without any loss
Actual	Next-stage instrument draws current → causes voltage drop → loading effect

▲ Loading effect is a **major cause of systematic error** in measurement systems.

1.2 Thévenin Equivalent Circuit

Simplify complex sensor internals into: **Ideal Voltage Source (E_{Th}) + Series Internal Impedance (Z_{Th})**.

$$V_L = E_{Th} \cdot \frac{Z_L}{Z_{Th} + Z_L}$$

Symbol	Meaning
V_L	Actual measured load voltage
E_{Th}	Open-circuit voltage (the true signal)
Z_L	Input impedance of load (measuring instrument)
Z_{Th}	Equivalent internal impedance of sensor

✅ **Golden rule:** To make $V_L \approx E_{Th}$, ensure $Z_L \gg Z_{Th}$.

2. Potentiometer Loading Effects

2.1 Review of Potentiometric Sensor

Uses sliding contact to change resistance → displacement → voltage.

2.2 With Load (Relative displacement $x = d/d_T$)

Quantity	Formula
Open-circuit voltage	$E_{Th} = V_S x$
Equivalent internal resistance	$R_{Th} = R_P x(1 - x)$
Actual load voltage	$V_L = V_S x \cdot \frac{1}{\frac{R_P}{R_L} x(1 - x) + 1}$

2.3 Two Design Constraints

Constraint	Formula	Design Rule
Non-linear error	$\hat{N} \approx 15 \frac{R_P}{R_L} \%$	R_P should be as small as possible
Maximum power rating	$\frac{V_S^2}{R_P} \leq \hat{W}$	V_S limited by R_P and power rating $\hat{W}$

▲ **Trade-off:** Small R_P → better linearity but higher power consumption; Large V_S → higher sensitivity but may exceed power rating.

3. Random Signals — Statistical Characteristics

3.1 Deterministic vs. Random

Type	Description	Example
Deterministic	Future values accurately predictable	Sine waves, step signals
Random	Impossible to write a definitive time function	Industrial temperature fluctuations, EM noise

✅ **Strategy:** Give up on predicting exact values. Instead, study **statistical patterns**.

3.2 Core Statistics

Statistic	Formula	Meaning
Mean $\bar{y}$	$\frac{1}{N} \sum y_i$	DC offset

Statistic	Formula	Meaning
Standard deviation σ	$\sqrt{\frac{1}{N} \sum (y_i - \bar{y})^2}$	Fluctuation from mean
RMS y_{rms}	$\sqrt{\frac{1}{N} \sum y_i^2}$	Signal energy

✅ **Special case:** When mean = 0, $\sigma = y_{rms}$.

3.3 Power-Energy Relationship (1Ω reference)

$$W_{TOT} = y_{rms}^2 = \bar{y}^2 + \sigma^2$$

Total Power = DC Power + AC Power

4. Autocorrelation Function (ACF)

4.1 Concept

Physical meaning: Measures the degree of "similarity" between a signal at the current moment and itself after a time delay β .

Function: Acts like a "sieve" — filters out random chaotic noise to **highlight periodic patterns** hidden within the signal.

4.2 Discrete ACF Formula

$$R_{yy}(m\Delta T) = \frac{1}{N} \sum_{i=1}^N y_i y_{i-m}$$

Symbol	Meaning
N	Total number of sampling points
m	Number of delay steps ($m\Delta T$ = total delay time β)
y_i, y_{i-m}	Sampled values at current moment and m moments prior

4.3 Calculation Steps

1. Create a duplicate copy of the original signal
2. Shift the duplicate by m units
3. Multiply aligned values term-by-term
4. Calculate the average of these products

5. Noise Characteristics & SNR

5.1 Internal Noise

Source: Random thermal motion of electrons in resistors and semiconductors (Johnson-Nyquist / Thermal Noise).

Characteristics: White Noise — energy distributed uniformly across an extremely wide frequency range.

5.2 Quantifying White Noise

Quantity	Formula	Unit
PSD ϕ	Power per unit frequency	W/Hz
Total noise power	$W_N = \phi \times f_C$	W
Noise RMS voltage	$V_{N,rms} = \sqrt{W_N}$	V

where f_C is the system bandwidth (cutoff frequency).

✔ Since thermal noise mean = 0, $V_{N,rms} = \sigma_N$.

5.3 Signal-to-Noise Ratio (SNR)

$$SNR_{dB} = 10 \log_{10} \frac{W_S}{W_N} = 20 \log_{10} \frac{V_S}{V_N}$$

SNR	Meaning
> 0 dB	Signal stronger than noise
= 0 dB	Signal equals noise
< 0 dB	Noise exceeds signal → signal is submerged/lost

6. Interference & Coupling Mechanisms

6.1 Sources

Source	Description
50/60 Hz mains	Most common "hum" / "mains pick-up"
Motor switching	Transient pulses
RF transmitters	Radio frequency interference

6.2 Coupling Mechanisms

Mechanism	Physical Process
Inductive	Changing magnetic field (di/dt) cuts across measurement loop → induces interference voltage
Capacitive	Stray capacitance between HV cables and signal lines → AC voltage leaks into measurement circuit
Ground loops	If sensor and receiver grounded at different points → potential difference V_E → interference current i_E

6.3 Two Forms of Interference Damage

Type	Description	Severity
Series mode (Normal mode, V_{SM})	Interference superimposed in series with useful signal	Most critical — directly causes measurement error
Common mode (V_{CM})	Both signal lines rise simultaneously relative to ground	Indirect — but can convert to series mode

7. Noise and Interference Suppression

7.1 Cutting Off Entry Paths

Against	Method	Principle
Inductive coupling	Twisted pairs	Induced EMF in adjacent loops has opposite polarity → cancellation
Capacitive coupling	Electrostatic shielding	Metal enclosure acts as Faraday cage
Ground loops	Single-point grounding	Shield grounded at only one end → no ground loop current i_E
Common-mode	Differential amplifier	Amplifies only difference, rejects common-mode

$$V_{CM_error} = \frac{V_{CM}}{CMRR}$$

✓

Higher CMRR (Common-Mode Rejection Ratio) → less common-mode error.

7.2 Signal Processing Methods

Method	Application	Formula
Band-pass filtering	Remove noise outside signal bandwidth	—

Method	Application	Formula
Signal averaging	Periodically repeating signals	$\sigma_{AV} = \frac{\sigma}{\sqrt{p}}$

where p = number of averages.

✅ **Averaging principle:** Positive and negative random noise fluctuations cancel over multiple averages; consistent signal is preserved. SNR improves by $\sqrt{p}$.

8. Time Domain vs. Frequency Domain

8.1 The Analogy

Domain	Like a...
Time domain	Cooked dish — you taste the overall flavor (aggregate waveform)
Frequency domain	Recipe and ingredient list — tells you 10 g salt (LF signal), 0.5 g pepper (HF noise), random impurities (white noise)

8.2 Discrete Fourier Transform (DFT)

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

8.3 Fast Fourier Transform (FFT)

- An extremely efficient algorithm for computing the DFT
- In Python/MATLAB: `fft(y)` transforms thousands of time-domain points into a spectrum instantly

8.4 How to Read an FFT Spectrum

Feature	Meaning
Spikes (peaks)	Strong periodic signal at that specific frequency
Noise floor	Gaussian white noise — appears as "fuzzy grass" spread across the spectrum

9. Nyquist-Shannon Sampling Theorem

$$f_s > 2f_{max}$$

To perfectly reconstruct the original signal without distortion, the sampling frequency must be **strictly greater than twice** the highest frequency component in the signal.

Aliasing

If $f_s < 2f_{max}$ → high-frequency signals **disguise themselves** as low-frequency signals in the spectrum.

Analogy: A fast-rotating car wheel appearing to spin backward — visual aliasing from the eye's insufficient sampling rate.

10. Summary Table

Concept	Key Formula / Method
Loading effect	$V_L = E_{Th} \cdot Z_L / (Z_{Th} + Z_L)$; require $Z_L \gg Z_{Th}$
Potentiometer non-linearity	$\hat{N} \approx 15 \frac{R_p}{R_L} \%$
Total signal power	$W_{TOT} = y_{rms}^2 = \bar{y}^2 + \sigma^2$
Autocorrelation	$R_{yy}(m\Delta T) = \frac{1}{N} \sum y_i y_{i-m}$
Noise power	$W_N = \phi \times f_C$
SNR	$SNR_{dB} = 10 \log_{10}(W_S/W_N)$
Common-mode rejection	$V_{CM_error} = V_{CM}/CMRR$
Averaging noise reduction	$\sigma_{AV} = \sigma/\sqrt{p}$
Nyquist theorem	$f_s > 2f_{max}$

✅ **Key Takeaway:** Loading effects arise because every real instrument has finite Z_L — the cure is $Z_L \gg Z_{Th}$. Random noise is quantified via PSD (ϕ) and bandwidth (f_C): $W_N = \phi f_C$. The frequency domain (FFT) is the "dimensionality strike" that separates signal from noise by their frequency fingerprints. Nyquist's theorem is the unbreakable digital rule: sample at **more than twice** your highest frequency of interest.